

Rotation Matrix in 3D — from 2D, via the Z-axis first

1 Warm-up: 2D rotation *is* rotation about the z -axis

In 2D we derived the rotation matrix (counterclockwise by θ):

$$R_{2D}(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Now observe: a rotation *in the $\mathbb{R}^2$ plane* is exactly a rotation in $\mathbb{R}^3$ *about the z -axis*, where every point keeps its z -coordinate fixed. The axis of the 2D rotation is therefore the z -axis. This gives us the 2D matrix *embedded* into 3×3 : keep the 2×2 block, and append the z -axis entries so that z is untouched:

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Multiplying a point (x, y, z) :

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \\ z \end{pmatrix}.$$

So the x, y part spins by the 2D matrix, and $z' = z$: the **z -component is untouched**. This is the whole content of “2D rotation = rotation about z .”

Note: the z -axis is not “imaginary” — it is the perfectly real third direction that is simply *perpendicular to the page* in a 2D drawing. It points “out of the paper.”

2 The same recipe for all three axes

By the identical reasoning, rotation about y (angle ϕ) and about x (angle ψ) are got by keeping that axis’s row and column as the axis itself and putting the 2D block in the other two coordinates:

$$R_y(\phi) = \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix}, \quad R_x(\psi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & -\sin \psi \\ 0 & \sin \psi & \cos \psi \end{pmatrix}.$$

In each matrix the **pivot axis’s row and column** read like the axis vector itself (1 on the diagonal, 0 elsewhere), and the remaining 2×2 block is exactly the 2D rotation within the perpendicular plane. (The sign placement in R_y differs from R_z because the x - z plane is read with y pointing the other way; both blocks are ordinary 2D rotations in their planes.)

3 Order matters: the ZYX (intrinsic) Euler convention

A *general* 3D orientation is a composition of these single-axis rotations, and **order matters**: rotating about z then about y is not the same as the reverse. The standard aerospace / robotics convention is **intrinsic Z–Y–X** (body-axis “yaw–pitch–roll”):

1. **Yaw:** rotate about the body's z -axis by θ .
2. **Pitch:** rotate about the body's *new* y -axis by ϕ (the y -axis has *moved* after step 1).
3. **Roll:** rotate about the body's *new* x -axis by ψ (moved again after step 2).

Each subsequent rotation happens about the axis that has just been rotated — this is the “body frame” behavior shown in the video. The total rotation is the right-to-left product

$$R = R_x(\psi) R_y(\phi) R_z(\theta)$$

which is applied to a vector as $v' = Rv$; the rightmost matrix R_z acts first, then R_y on its result, then R_x , matching the intrinsic order.

Fixed vs rotating axes (the subtlety). If instead one rotates about the *fixed* world axes, the composite matrix comes out *reversed*:

$$\text{intrinsic } z, y, x \quad \Longleftrightarrow \quad \text{extrinsic } x, y, z.$$

That is, rotating about the moving y -then- x equals rotating about the fixed x -then- y in reverse order. The two readings describe the same final geometry; only the order of ψ, ϕ, θ in the product interpretation changes.

4 Summary

- 3D rotation about any single axis = the 2D rotation in the perpendicular plane, with that axis left untouched.
- R_z is the very 2D matrix we derived, plus the 0,0,1 row/column.
- A general 3D orientation is an *ordered* product of single-axis ones; the ZYX intrinsic convention applies z , then y , then x , about the already-rotated axes.